

Lesson 1 · Gradients, Differentiation and Tangents

Learning intention

Differentiate polynomials, and use $f'(x)$ to find the gradient of a curve and the equation of a tangent.

Success looks like

- I can differentiate any polynomial, expanding brackets first.
- I can choose: gradient at a point, point with a gradient, or unknown constant.
- I can write the equation of a tangent in three steps.

Slide 1 · Lesson 1 · Gradients, Differentiation and Tangents

Teaching line, not a question slide.

Say: calculus is about gradients of curves. Today is the toolkit every later lesson uses.

Timing: Do Now 10 min, three teach-and-practise rounds about 20 min each, practice sheet 30 min, wrap-up 5 min.

Do Now

- 1 Find the gradient of the line through (1, 3) and (4, 12).
- 2 Expand and simplify $3x(x - 4)$.

- 3 Find the equation of the line with gradient -2 that passes through (3, 5).
- 4 Solve $x^2 - 5x + 6 = 0$.

Slide 2 · Do Now

Q1 3

Q2 $3x^2 - 12x$

Q3 $y = -2x + 11$

Q4 $x = 2$ or $x = 3$

Short solution:

Q1 $m = \frac{12 - 3}{4 - 1} = \frac{9}{3} = 3$

Q2 $3x \times x - 3x \times 4$

Q3 $y - 5 = -2(x - 3)$, so $y = -2x + 6 + 5$

Q4 $(x - 2)(x - 3) = 0$

Watch for:

Q1 run over rise gives $\frac{1}{3}$

Q3 sign error when expanding $-2(x - 3)$

Diagnostic:

Q1 wrong: gradient idea is shaky; spend longer on the one-idea slide

Q2 wrong: expect L1 errors (expanding before differentiating)

Q3 wrong: the tangent step (round 3) will fail; re-teach $y - y_1 = m(x - x_1)$ first

Q4 wrong: allow the calculator EQUA menu for quadratics today

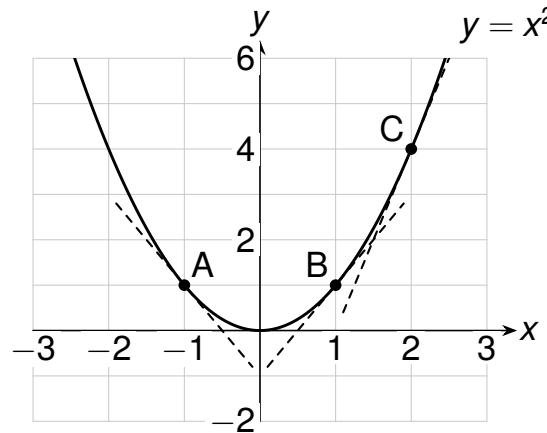
The one idea

A curve's steepness changes from point to point.

The gradient of a curve at a point = the gradient of its **tangent** there.

$f'(x)$ is a **gradient machine**: put in x , get out the gradient.

At A, B and C: going up or down?
Which is steepest?



Slide 3 · The one idea

A down, gradient -2

B up, gradient 2

C up and steepest, gradient 4

Short solution:

for $y = x^2$ the gradient machine is $f'(x) = 2x$: $2(-1) = -2$, $2(1) = 2$, $2(2) = 4$

Watch for:

students reading steepness from height; C is steeper because the tangent rises faster, not because it is higher

The power rule

$$\frac{d}{dx}(ax^n) = an x^{n-1}$$

Multiply by the power, then take 1 off the power. Do each term on its own.

term	x^5	$4x^3$	$-6x^2$	$7x$	9
derivative					

Expand brackets first: the rule works on single terms only.

Slide 4 · The power rule

$$x^5 \rightarrow 5x^4$$

$$4x^3 \rightarrow 12x^2$$

$$-6x^2 \rightarrow -12x$$

$$7x \rightarrow 7$$

$$9 \rightarrow 0$$

Short solution:

$$7x = 7x^1 \rightarrow 7 \times 1 \times x^0 = 7; \text{ a constant is a flat line, gradient } 0$$

Watch for:

$$7x \rightarrow 7x \text{ or } 7x \rightarrow 0 \text{ (L3)}$$

$$\text{the sign travels with the term: } -6x^2 \rightarrow -12x$$

Worked examples · differentiate

1 Differentiate $f(x) = 3x^4 - 5x^2 + 7x - 2$.

Step 1: term by term Step 2: $f'(x) = \dots$

2 Differentiate $y = (2x + 1)(x - 3)$.

Step 1: expand Step 2: $\frac{dy}{dx} = \dots$

Slide 5 · Worked examples · differentiate

Q1 $f'(x) = 12x^3 - 10x + 7$

Q2 $\frac{dy}{dx} = 4x - 5$

Short solution:

Q1 $3x^4 \rightarrow 12x^3$, $-5x^2 \rightarrow -10x$, $7x \rightarrow 7$, $-2 \rightarrow 0$

Q2 $y = 2x^2 - 6x + x - 3 = 2x^2 - 5x - 3$

Watch for:

Q2 differentiating each bracket and multiplying gives $2 \times 1 = 2$ (L1)

Practice

Differentiate.

1 $f(x) = 5x^3 - 4x^2 + x - 9$

2 $y = x(x^2 + 6)$

3 $y = (x - 2)^2$

Slide 6 · Practice

Q1 $f'(x) = 15x^2 - 8x + 1$

Q2 $\frac{dy}{dx} = 3x^2 + 6$

Q3 $\frac{dy}{dx} = 2x - 4$

Short solution:

Q2 expand: $y = x^3 + 6x$

Q3 expand: $(x - 2)^2 = x^2 - 4x + 4$

Watch for:

Q3 $(x - 2)^2 \neq x^2 + 4$; the middle term $-4x$ is the one that matters

Q1 the x -term gives 1, not 0

Three question types

given $\rightarrow$ asked	route
$x \rightarrow$ gradient	differentiate, then substitute x into $f'(x)$
gradient $\rightarrow$ point	solve $f'(x) = \text{gradient}$ for x ; then y from $f(x)$
gradient at $x \rightarrow$ constant	differentiate with the letter in; substitute x ; solve

The **gradient** comes from $f'(x)$. The y -coordinate comes from $f(x)$.

Slide 7 · Three question types

Teaching line, not a question slide.
Ask the student to underline what is given and circle what is asked before choosing a row.
L2 (substituting into the wrong function) is the most common slip in this lesson.

Worked examples · gradients

3 Find the gradient of $f(x) = x^3 - 6x^2 + 4$ at the point where $x = 5$.

4 Find the point on $y = x^2 - 7x + 1$ where the gradient is 3.

5 The curve $f(x) = ax^2 + 3x - 1$ has gradient 11 at $x = 2$. Find a .

Slide 8 · Worked examples · gradients

Q3 15

Q4 (5, -9)

Q5 $a = 2$

Short solution:

Q3 $f'(x) = 3x^2 - 12x$; $f'(5) = 75 - 60$

Q4 $2x - 7 = 3$ gives $x = 5$; $y = 25 - 35 + 1$

Q5 $f'(x) = 2ax + 3$; $4a + 3 = 11$

Watch for:

Q4 stopping at $x = 5$: a point needs y as well

Q5 differentiating a away; a is a number and stays

Practice

- 1 Find the gradient of $f(x) = 2x^3 + x^2 - 4x$ at the point where $x = 2$.
- 2 Find the coordinates of the point on $y = x^2 + 4x - 3$ where the gradient is 10.
- 3 The curve $f(x) = kx^3 - 2x$ has gradient 22 when $x = 2$. Find k .

Slide 9 · Practice

Q1 24

Q2 (3, 18)

Q3 $k = 2$

Short solution:

Q1 $f'(x) = 6x^2 + 2x - 4$; $f'(2) = 24 + 4 - 4$

Q2 $2x + 4 = 10$ gives $x = 3$; $y = 9 + 12 - 3$

Q3 $f'(x) = 3kx^2 - 2$; $12k - 2 = 22$

Watch for:

Q1 substituting into $f(x)$ gives 16 (L2)

Q3 $3kx^2$ at $x = 2$ is $12k$, not $6k$

Equation of a tangent

Write it in this order

1. **Point:** y_1 from $f(x)$.
 2. **Gradient:** $m = f'(x_1)$, a number.
 3. Line: $y - y_1 = m(x - x_1)$, then $y = mx + c$.
- 6 Find the equation of the tangent to $y = x^3 - 2x + 1$ at $x = 2$.

Slide 10 · Equation of a tangent

Q6 $y = 10x - 15$

Short solution:

Q6 point $(2, 5)$; $\frac{dy}{dx} = 3x^2 - 2$, $m = 10$; $y - 5 = 10(x - 2)$

Watch for:

using $f'(x)$ for y_1 gives $(2, 10)$ (L2)

writing $y - 5 = (3x^2 - 2)(x - 2)$: the gradient must be a number (L4)

Worked example · two unknowns

7 The curve $y = ax^2 + bx$ passes through $(1, 5)$, and its gradient at that point is 8. Find a and b .

one equation from the **point** + one from the **gradient**

Parallel lines have equal gradients:

“tangent parallel to $y = 2x + 1$ ” means $f'(x) = 2$.

Slide 11 · Worked example · two unknowns

Q7 $a = 3, b = 2$

Short solution:

Q7 point: $a + b = 5$; gradient: $2a + b = 8$; subtract: $a = 3$

Watch for:

Q7 using only one condition and guessing the other letter

Merit marker: two linked conditions, clearly labelled

Practice

- 1 Find the equation of the tangent to $f(x) = 2x^2 - 3x + 4$ at $x = 2$.
- 2 The tangent to $y = x^2 - 4x + 7$ at the point P is parallel to the line $y = 2x + 1$. Find the equation of this tangent.
- 3 The tangent to $y = x^3$ at $x = 1$ meets the curve again. Find the coordinates of this second point.

Slide 12 · Practice

Q1 $y = 5x - 4$

Q2 $y = 2x - 2$

Q3 $(-2, -8)$

Short solution:

Q1 point $(2, 6)$; $f'(x) = 4x - 3$, $m = 5$; $y - 6 = 5(x - 2)$

Q2 $2x - 4 = 2$ gives $x = 3$; point $(3, 4)$; $y - 4 = 2(x - 3)$

Q3 tangent $y = 3x - 2$; $x^3 - 3x + 2 = 0$; $(x - 1)^2(x + 2) = 0$, so $x = -2$

Watch for:

Q2 the point P is not given: find it from the gradient first (Merit)

Q3 Excellence stretch; the repeated root $x = 1$ is the point of contact

Ways to lose marks

L1 Not expanding brackets before differentiating.

L2 Gradient from $f'(x)$; y -coordinate from $f(x)$.

L3 $7x \rightarrow 7$ and a constant $\rightarrow 0$.

L4 The tangent's gradient must be a number, not $f'(x)$.

L6 "Use calculus": write $f'(x)$ down.

Slide 13 · Ways to lose marks

Teaching line, not a question slide.

Full list L1 to L6 is in the handout; L5 (brackets around negative x) is there too.

Ask: which of these did you do today? Tick them in the handout.

Summary

1. $\frac{d}{dx}(ax^n) = anx^{n-1}$; expand brackets first.
2. Gradient at a point: substitute x into $f'(x)$.
3. Given a gradient: solve $f'(x) = m$; y from $f(x)$.
4. Tangent: point, gradient, then $y - y_1 = m(x - x_1)$.

Slide 14 · Summary

Teaching line, not a question slide.

Ask the student to say each line back before looking at the screen.

Practice sheet · 30 minutes

Work alone on the practice sheet.

- Questions 1 to 4: warm-up, about 8 minutes.
- Questions 5 to 10: exam questions, about 22 minutes.
- Show every line of working; write $f'(x)$ each time.

Stuck for more than 3 minutes? Move on and come back.

Slide 15 · Practice sheet · 30 minutes

Q1 $f'(x) = 24x^3 - 9x^2 + 2$

Q2 11

Q3 $(3, -7)$

Q4 $y = 9x - 25$

Q5 89

Q6 $y = 6.5x - 8$

Q7 $c = 4$

Q8 204

Q9 $p = -4, q = 0.5$

Q10 $y = -2x - 2$

Short solution:

Q6 point $(2, 5)$; $f'(x) = 1.5x^2 + 0.5, m = 6.5$

Q7 $f'(x) = 3 + 2cx - 6x^2$; $f'(2) = 4c - 21 = -5$

Q9 point: $2p - 4q = -10$; gradient: $p - 4q = -6$; subtract: $p = -4$

Q10 expand $y = 2x^2 - 6x$; point $(1, -4)$; $m = 4 - 6 = -2$

Watch for:

Q6 halves: keep $\frac{1}{2}x^3 \rightarrow \frac{3}{2}x^2$

Q10 must expand first (L1)

Full working: practice answer version (tutor's working, not an official mark scheme)

Homework

Homework sheet: 15 questions, about 60 minutes

- Questions 1 to 12: Achieved practice.
- Questions 13 and 14: Merit.
- Question 15: Extension (Excellence).

Optional extra: StudyTime Guide, “Stop and Check” on pages 10 and 12.

Slide 16 · Homework

Teaching line, not a question slide.

Answers: homework answer version (tutor’s working, not an official mark scheme).

Next lesson Do Now checks Q2 style (differentiate after expanding) and gradient at a point.